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Where to Start a Dynamic Skill: Learning Object Spinning from Contact Transitions

Anonymous Author(s)[1]

Abstract— **Learning dynamic in-hand skills requires initial grasps that allow motion to develop. Wrench-space analysis of pen spinning shows that closure is intermittent and initial static margins poorly predict sustained motion. We propose closure-event grasp discovery, which selects training starts when opposing fingertip contacts form during rotation. An angle-space certificate detects these transitions; persistence and future-survival checks retain configurations for zero-velocity resets of a fresh policy. The same certificate provides lightweight, rotation-gated reward shaping without online wrench-hull construction. On the original grasp cache, event-trained pen policies increase mean simulation sustain from 18% to 78% at equal second-stage training budget. Comparisons with matched non-event states and alternative initialization strategies support the value of contact-transition selection. The benefit persists without second-stage shaping, and useful resets can be recovered even from a poorly performing source. Object-specific training on ten simulated objects and deployment on eight physical objects further evaluate the shaping pipeline. These results identify contact timing as a useful principle for choosing where to begin learning dynamic manipulation.**

I. Introduction

Where should a dexterous hand start a dynamic skill? Classical grasp synthesis favors contacts that resist disturbances, quantified by force closure and the grasp wrench space (GWS) [1]–[3]. Object spinning poses a different requirement: the initial grasp must allow the hand to enter a sequence of releases and catches. A large static closure margin need not identify such a state, even when it indicates strong instantaneous wrench resistance.

The reset distribution determines which contact states a learner encounters. A grasp that survives settling may still hinder rotation, while a catch configuration may provide entry into a repeating gait. We therefore ask whether a grasp supports learning sustained motion as well as resisting immediate disturbance.

We analyze learned spinning policies for a 22-DoF hand. The unshaped pen teacher has fingertip force closure during only 29% of steps. Initial margins correlate weakly with episode length, and low instantaneous margins poorly predict drops because normal releases also lose closure. Margin decay is more informative; resolving it by wrench axis distinguishes loss of support from loss of attitude (Section IV).

Our main method, *closure-event grasp discovery* (Fig. 1), harvests transitions into an opposing fingertip pair during rotation. Persistence and future-survival checks select catches whose configurations, with velocities cleared, become fresh-policy resets. Selection uses an angle-space certificate derived from the planar antipodal-grasp condition [4], [5]. The certificate also supplies rotation-gated shaping, weighted to balance support and attitude failures. It thus both encourages the source to catch and identifies training starts. Evaluation on the original cache tests transfer beyond the discovered configurations.

On the pen, event resets raise average sustain on the standard grasp cache from 18% to 78% at equal second-stage budget; without second-stage shaping, the result is 5% versus 71%. Random source-rollout states reach 35% without survival filtering and 57% with it; non-event states matched in motion, critic value, contact count and future survival reach 65%. Jump-Start RL reaches 40%, and perturbing sweep starts with joint and pose noise remains at 18%. These controls support a benefit from transition selection over survival-filtered random resets and the tested forms of source guidance and start diversity. Separately, our shaping reaches 88% in a reward ablation, versus 15% for terminal quality and 25% for contact-count rewards. We also compare reset rankings, account for source-training cost, and evaluate object-specific shaping on ten simulated and eight physical objects (Section VII).

Our contributions are: (1) *closure-event grasp discovery*, which filters source-policy catches by persistence and future survival for zero-velocity fresh-policy resets; (2) per-step wrench-space analysis showing intermittent closure and weak predictive value of initial static margins; (3) an angle-space certificate used for rotation-gated shaping without online wrench-hull construction, with an empirical weight rule; and (4) evaluation on ten simulated and eight physical objects. Analysis tools, caches and logs will be released with the paper.

II. Related Work

Closure and grasp quality. The GWS represents the wrenches a contact set can resist [2], [5]; its largest origin-centered inscribed ball defines the Ferrari–Canny margin [1], [3]. These quantities describe a fixed contact geometry and its available force combinations. Two frictional contacts achieve planar force closure when their connecting segment lies strictly inside both friction cones [4]. We express this classical condition in angle space for a circular cross-section. The resulting pair is an antipodal pinch; w_{cage} retains the implementation's name and does not denote geometric caging. Our event criterion additionally requires motion and subsequent survival, which a positive static closure margin alone cannot certify.

Closure in learning. Differentiable closure measures support grasp synthesis [6]–[8]. Policy objectives include grasp-

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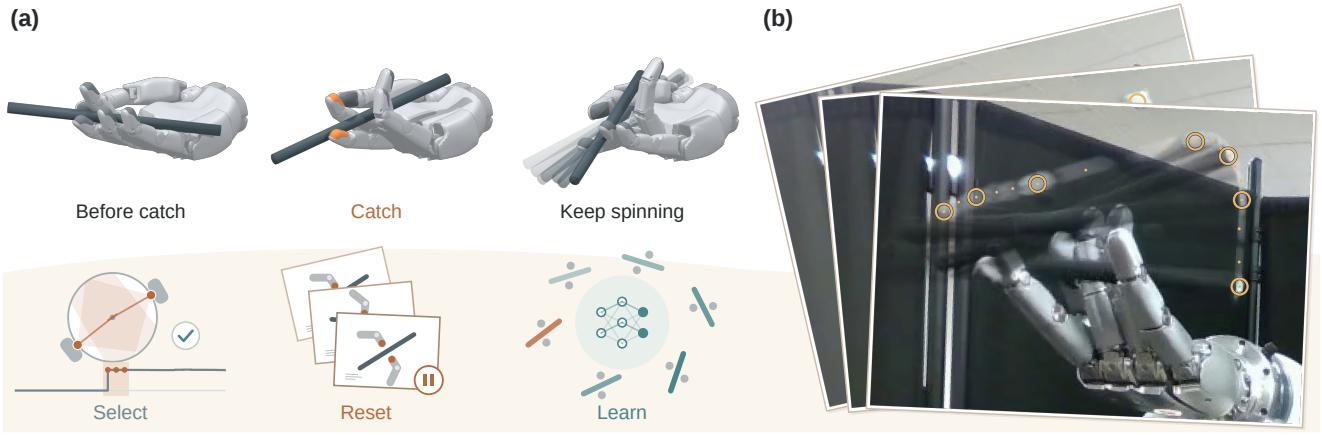


Fig. 1. (a) Catch-based resets and fresh learning, with method schematics below. (b) Recorded pen-motion composite with tip markers.

quality rewards [9], terminal Q_1 rewards [10], dense quality shaping for rolling [11], and demonstration-derived contact-wrench guidance [12]. In spinning, maintaining grasp quality must be reconciled with releasing fingers to continue the gait. Table I compares reward-level adaptations of dense quality and terminal Q_1, plus a contact-count control, with our motion-conditioned pair score. The dense baseline uses our closure magnitude; the cited methods motivate the reward family and are not reproduced end to end. Wrench-hull analysis remains offline and also motivates selecting training starts at contact transitions.

Dynamic in-hand manipulation. Rolling and finger gaiting involve contact-mode changes [13]; learned systems acquire object rotation [14], [15] and pen spinning [16]. Qi et al. adapt to object properties from proprioceptive history, while Wang et al. combine simulated trajectories with real-world replay and adaptation for pen-like objects. We retain the teacher–student structure of Wang et al. and study how contact geometry can improve reward design and training starts. Our object-specific shaping experiments assess this component across shapes; they do not establish that event discovery generalizes to unseen objects.

Reset distributions. DeepMimic's reference state initialization (RSI) samples states along reference motions [17]; reverse curricula expand starts outward from a goal [18]. R×R supplies resets through planning [19]; other approaches use preceding sub-policies [20] or restart archives [21]. Jump-Start RL uses a guide policy for a decreasing episode prefix before the learner takes over [22]. Continuous spinning has no single terminal configuration for a reverse curriculum. We compare unfiltered and survival-filtered random resets, Jump-Start guidance and noisy sweep starts (Table II). Our unfiltered control adapts RSI to source-rollout configurations with velocities cleared. These controls test trajectory reuse, survival filtering, source guidance and start perturbation.

III. Task, Policy, and Wrench-Space Instrumentation

A. Task and learning setup

Task and policy. A Sharpa Wave hand with 22 actuated joints and five elastomer fingertips spins objects in Isaac Lab, the successor to Orbit [23]. Physics runs at 2.5 ms with decimation 20, giving a 20 Hz policy and 400-step (20 s) episodes. Leaving the workspace or violating pose bounds terminates an episode. PPO [24] trains teachers in 8192 environments with an asymmetric critic. The actor receives a 132-D proprioceptive history, a 37-D privileged object state and a 100-point surface cloud. Its 22-D action increments joint position targets by up to 0.0417 rad per step under torque-limited PD control. Randomization covers mass, friction, restitution, joint noise and pushes.

Reward. The rotation reward uses net orientation change over a control step to suppress jitter. For elongated objects it measures the signed yaw rate of the long axis (*twirl*), excluding rolling about that axis; compact objects use the net angular rate about world $+z$. Object motion, pose drift, torque and work penalties complete the base reward. Shaped and unshaped runs share these settings and differ by the certificate term of Section V-B.

Grasp caches. Object-specific sweeps place objects around the fingertips at candidate gait phases. Candidates are settled, pushed and replayed twice; survivors are re-verified in a fresh process and filtered by pose bounds. For the reference pen (226 mm long, 11.6 mm diameter), four sweeps yield 9952 candidates and 1004 verified rows; removing initial tilts above 30° leaves the 897-row *standard cache*.

Evaluation. Let L be the recorded episode length. Simulation *sustain* is $P(L \geq 399 \mid L \geq 5)$ in completed evaluation episodes; the five-step condition excludes immediate initialization failures. *Pinned survival* is $P(L \geq 390)$ over all cache rows and repeated resets, retaining the fixed evaluator's near-timeout threshold. These protocols are compared only within themselves. We report median task revolutions unless a mean is explicitly stated. Hardware uses the separate Sustain@20s protocol in Section VII-B. The reference teachers are *pinch3* (unshaped, 16 h), *cage01* ($w_{\text{cage}} = 0.1$, 4 h) and *cage005* ($w_{\text{cage}} = 0.05$, 4 h).

B. Wrench-space quantities

Contact model. Pen analysis models the lateral surface as a finite cylinder and excludes end-cap contacts. Duty-cycle measurement and harvesting accept an elastomer center

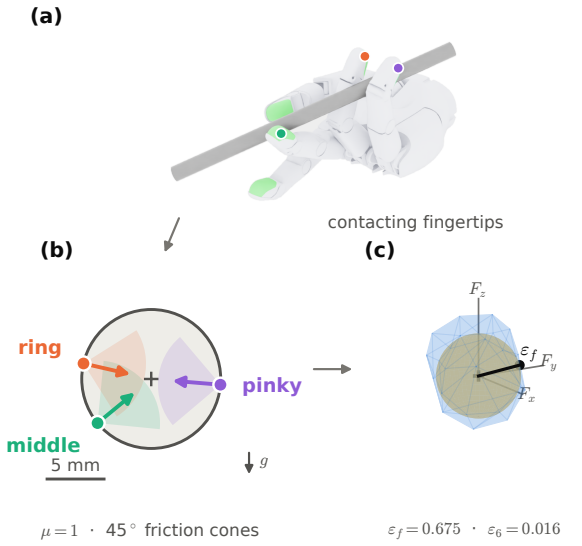


Fig. 2. Wrench-space instrumentation of an unshaped-teacher step. (a) Three fingertips pass the contact test. (b) Estimated contact points, inward normals and friction cones on the pen cross-section (diameter 11.6 mm). (c) Force-subspace hull and its inscribed ball: $\varepsilon_f = 0.675$, versus $\varepsilon_6 = 0.016$ for the torsion-limited 6-D hull.

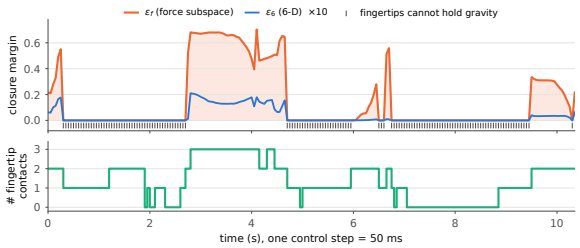


Fig. 3. Closure over a reference-pen rollout of the unshaped teacher. Top: force-only margin ε_f and six-dimensional margin ε_6 (scaled by 10); black ticks mark steps at which the fingertip gravity-balance program is infeasible. Bottom: detected fingertip contacts. Contact count and wrench resistance vary through the gait, with prolonged intervals of zero fingertip closure between pinches. The plot describes the fingertip model; other hand surfaces may still support the object.

within 11 mm of the surface, or within 22 mm with incoming joint force above 0.02 N. Static-cache and scalar drop analyses use the 11 mm geometric test alone; the directional analysis uses 13 mm with the same force fallback within twice that distance. The elastomer origin lies 7–9 mm inside the pad, motivating these proximity approximations. Nearest surface points and inward radial normals define eight-edge friction pyramids with $\mu = 1$, within the randomized range $[0.3, 3.0]$. Two torsional primitives per pad approximate soft-finger friction with $\gamma = 0.005\,\mathrm{m}$ [25]. A unit primitive force f at center-of-mass offset r gives $w = [f; (r \times f)/\rho]$, with ρ the pen half-length.

Closure and direction. For the primitive-wrench hull $\mathcal{H}$, ε_6 is the radius of its largest origin-centered inscribed ball; ε_f uses only force components. Margins are set to zero when the origin is outside the hull. A gravity-balance linear program measures the minimum primitive activation needed to hold the pen. The model covers fingertip pads only: zero margin does not exclude support from a phalanx or the palm.

To compute the margins, write the hull as $\mathcal{H} = \{w : a_j^\top w + b_j \le 0,\ j = 1, \ldots, J\}$, with unit facet normals $\|a_j\|_2 = 1$. When the origin is strictly inside the hull,

$$\varepsilon_6 = \min_j(-b_j), \qquad m_d = \min_{j:a_j^\top d>0} \frac{-b_j}{a_j^\top d}. \tag{1}$$

Here d is a unit wrench direction: m_d is the radial distance from the origin to the boundary along that ray. Both quantities are set to zero without closure. The force-only margin repeats the inscribed-ball calculation on the 3-D force hull. Normalizing torque by ρ places force and torque coordinates on a common scale, so numerical margins should be interpreted with this normalization fixed.

We query gravity support $+F_z$, lateral force (minimum over $\pm F_x, \pm F_y$), tilt torque ($\pm T_x, \pm T_y$) and task torque ($\pm T_z$), all in world coordinates. Taking the minimum over signed axes measures resistance in both senses; the gravity query is one-sided because the required support force points upward. These directional capacities retain the same contact model and wrench primitives as the scalar hull, making their temporal changes directly interpretable within an analysis protocol.

Temporal signals. The closure duty cycle is the fraction of steps with $\varepsilon_6 > 0$ above numerical tolerance. The scalar decay score is $s_t = -[\varepsilon_f(t) - \varepsilon_f(t-K)]\mathbf{1}[\varepsilon_f(t-K) > h]$, with $K = 10$ steps (0.5 s) and $h = 0.2$. Directional analysis analogously compares changes in m_d. For instantaneous-margin AUCs, lower margin means higher predicted risk. All hull and linear-program quantities are computed offline.

IV. What Closure Reveals During Spinning

We examine whether static grasp quality describes a spinning gait, selects its starting states or anticipates failure.

A. Closure is intermittent

Under *pinch3*, fingertip force closure holds for $D = 0.29$ of 12,720 steps across 50 pushed episodes (per-episode median 0.34, mean 1.27 contacts per step). A representative rollout never exceeds $\varepsilon_6 = 0.02$, reflecting weak torsional resistance. Under *cage01*, the duty cycle rises to $D = 0.82$ across 249 episodes (median 0.93, mean 2.40 contacts). Thus closure frequency depends on the policy; a positive-margin reward provides little signal during much of the unshaped gait.

Fig. 3 shows why a single contact statistic is insufficient. Some intervals contain a fingertip contact but no positive closure margin, whereas a regrasp can sharply increase the force margin while the six-dimensional margin remains small. The gravity-effort feasibility test addresses yet another question: whether the modeled fingertips can balance the current weight. Recording these signals together distinguishes contact presence, omnidirectional resistance and support feasibility before interpreting an apparent loss of closure as failure.

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B. Initial closure weakly predicts persistence

Across 897 initial grasps, each rolled out once under *pinch3*, initial ε_6 has Spearman $\rho = 0.07$ with episode length ($p = 0.03$), and ε_f has $\rho = 0.06$ ($p = 0.09$). Splitting at median ε_f changes median lifetime from 252 to 253 steps. Under ten pinned repeats with *cage01*, the 462 initially certified rows spin more than the remaining rows (mean 1.48 versus 1.24 revolutions, Mann–Whitney $p = 8\times10^{-7}$), while their survival difference is small (0.70 versus 0.68, $p = 0.2$). Survival spans only 0.69–0.72 across margin quintiles within those rows.

Placement carries more predictive information: proximal offset from the index–middle fingertip midpoint gives AUC 0.69 for whether at least half a row's repeats survive. A cross-fitted five-feature placement model reaches 0.70 versus 0.61 for six static wrench features. Static margins alone therefore provide a weak reset-selection rule in this setting.

C. Margin decay precedes failure

For 50 pushed episodes per teacher (45 drops for *pinch3*, 12 for *cage01*), positive prediction labels mark steps within ten control intervals of a terminal drop, including the final recorded step. Drop-aligned traces show an elevated-closure phase followed by decay toward zero. A low instantaneous ε_f is anti-predictive for *pinch3* (AUC 0.29), whereas the decay score reaches AUC 0.82 and 0.81 for the two teachers (95% episode-bootstrap intervals $[0.78, 0.87]$ and $[0.75, 0.87]$). These associations motivate a temporal diagnostic; they do not establish that strong closure causes drops.

D. Directional decay distinguishes failure channels

We compare the last pinch before a normal release with that before a terminal drop. Under *pinch3*, gravity-margin decay separates them with AUC 0.86 versus 0.71 for scalar ε_6 decay. In the larger *cage01* set (249 episodes, 48 eligible pre-drop pinches and 177 releases), gravity is less discriminative. A separate lead-time analysis against release states reaches AUC 0.89 for tilt decay at 0.1 s before terminal drops, matching scalar decay at that horizon. The weakest axis alone is uninformative: tilt torque limits 70–96% of pinches. These observations suggest encouraging opposing contacts during motion, while monitoring both support and attitude failures when selecting the shaping weight.

V. Closure-Structured Shaping

A. The angle-space closure certificate

Definition 1 (closure certificate): For a contact set $\mathcal{C}_t$ with inward unit normals $\hat{n}_i$, define

$$c_t = \max_{\substack{i,j\in\mathcal{C}_t \\ i<j}} \max(0, -\hat{n}_i \cdot \hat{n}_j) \in [0, 1], \tag{2}$$

with $c_t = 0$ when fewer than two fingertips touch.

Proposition 1 (planar pair closure): Consider two distinct contacts on a circular cross-section, with inward radial normals $\hat{n}_i, \hat{n}_j$ and Coulomb coefficient $\mu > 0$. Their planar point-contact grasp resists arbitrary in-plane force and torque about the cylinder axis if and only if

$$-\hat{n}_i \cdot \hat{n}_j > \kappa(\mu) := \frac{1-\mu^2}{1+\mu^2}. \tag{3}$$

Proof. The chord connecting contacts must lie strictly inside both friction cones [4], [5]. For angular separation $\theta \in (0, \pi]$, it makes angle $(\pi - \theta)/2$ with each inward normal. Hence closure requires $\theta > \pi - 2\arctan\mu$, equivalent to $-\cos\theta > \cos(2\arctan\mu) = \kappa(\mu)$. Equality is marginal. □

For $\mu = 1$, $c_t > 0$ certifies planar closure of at least one pair. More generally, $c_t > \max(0, \kappa(\mu))$ is exact for pair closure when $\mu \le 1$ and sufficient when $\mu > 1$; at $\mu = 0.3$, the threshold is 0.835. The score itself is dense above an obtuse normal separation and reaches one for a single antipodal pair. Additional contacts can improve the best pair before saturation, but contact count earns no separate bonus. This planar certificate does not guarantee full 6-D closure, resistance to long-axis tilt, or dynamic stability.

B. Reward and implementation

We add the bounded score only during positive task rotation:

$$r_t = r_t^{\text{task}} + w_{\text{cage}}\, c_t\, \mathbf{1}[\omega_t > 0.1\,\text{rad/s}], \tag{4}$$

where r_t^{task} includes the base regularization and ω_t is the signed task rate of Section III. The gate prevents a stationary hold from earning the bonus. The term applies at full weight from the first update and requires only pairwise dot products, without a hull computation.

Online reward contacts use the 11 mm proximity test on the object's bounding cylinder, with distances measured from its clamped axis segment. Offline diagnostics and pen harvesting additionally use the force-assisted lateral-contact rule of Section III-B. Both apply (2), but their contact sets need not coincide. The reward does not adapt its score to the randomized friction coefficient; its closure interpretation is nominal at $\mu = 1$.

TABLE I

Reward-formulation ablation on the reference pen

reward	sustain (%)	rev.	closure duty
base	2	0.4	0.29
dense quality [9], [11]	19	0.6	**0.94**
gated closure magnitude	48	1.1	0.88
terminal Q_1 [10]	15	0.6	0.50
contact-count reward	25	0.8	0.70
certificate, no gate	35	0.8	0.86
ours	**88**	**2.1**	0.84

Base, dense/gated magnitude, certificate and ours: three-seed means.

Table I compares reward formulations added to the same base task objective. Dense quality is our per-step closure-magnitude adaptation of grasp-quality rewards [9], [11]; terminal Q_1 adapts the end-of-episode quality bonus of She et al. [10], paying it only once. Contact-count rewards the number of touching fingertips. These are reward-level adaptations to spinning. Terminal Q_1 reaches 15% sustain, 0.6 revolutions and 0.50 closure duty; contact-count reaches

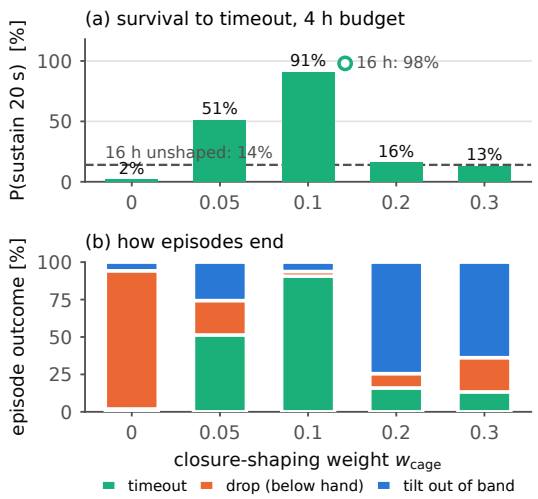


Fig. 4. Shaping-weight sweep with the real-calibrated hand model. (a) Sustain after 4 h; dashed line: 16 h unshaped teacher; hollow marker: $w_{\text{cage}} = 0.1$ continued to 16 h. (b) Termination fractions over all completed episodes.

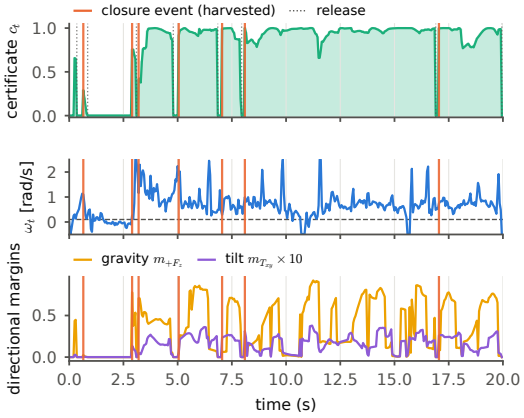


Fig. 5. One 20 s source rollout (cache row 94). Orange lines mark harvested closure onsets; dotted lines mark releases. Panels show c_t, task rate ω_t, and gravity/tilt margins (tilt scaled by 10). The displayed rate is clipped to $[-0.5, 2.5]$ rad/s.

25%, 0.8 and 0.70, versus 88%, 2.1 and 0.84 for ours. Dense magnitude has the highest closure duty but low sustain; gating or the certificate alone also falls below their combination. The results favor motion-conditioned pair geometry over terminal quality or contact count alone. This ablation is separate from the named-teacher dose sweep below.

C. Dose response and weight selection

Fig. 4 holds the training recipe and cache fixed and allocates 4 h per weight. Sustain rises from 2% at $w_{\text{cage}} = 0$ to 51% at 0.05 and 91% at 0.1, then falls to 16% at 0.2 and 13% at 0.3. The corresponding median twirls are 1.03, 2.58, 2.14, 0.89 and 0.48 revolutions. The 16 h unshaped teacher reaches 14% sustain and 1.47 revolutions; continuing the $w_{\text{cage}} = 0.1$ run to 16 h reaches 98% and 2.00.

Termination channels explain the weight choice. Without shaping, 92% of completed episodes end below the hand. At $w_{\text{cage}} \geq 0.2$, 64–74% violate the tilt band. At $w_{\text{cage}} = 0.1$, 91% reach timeout, with both failures reduced. Our empirical procedure is to diagnose the decaying wrench axis, add motion-gated pair shaping, then select a weight that reduces support failures without increasing attitude failures. It minimizes the observed failure tradeoff; neither channel is guaranteed to vanish.

Distillation. DAgger [26] produces a proprioception-only student of the shaped teacher with median length 348 steps, 48% sustain and 0.74 twirl revolutions, without student RL. The unshaped teacher's student reaches 209 steps and 33% after 6 h of PPO fine-tuning. These single-student results use different fine-tuning budgets.

VI. Closure-Event Grasp Discovery

The reset state of a dynamic skill must permit motion to resume. We therefore use a source policy's contact transitions to discover candidate configurations, then evaluate them both as fixed-policy starts and as training resets.

A. Event harvesting

Given a source policy and an initial cache, we roll out with training pushes at nominal object parameters, without observation/action noise, and log joint state, object pose and fingertip positions. Offline instrumentation gives the contact set, c_t and ω_t. A candidate onset satisfies

$$c_{t-1} = 0, \quad \min_{s=t}^{t+2} c_s > 0, \quad \omega_t > 0.1\,\text{rad/s}. \tag{5}$$

Thus closure persists for three sampled steps; the motion test applies at onset. We retain a candidate if the episode avoids a drop for the next 5 s, or reaches its scheduled timeout sooner. This uses future rollout information during cache construction.

Each reset stores the event's joint positions as both positions and PD targets, together with object pose. All velocities are zeroed: the reset reuses a configuration observed during motion, not the event's complete dynamic state. From 249 *cage01* rollouts, 561 of 568 onsets pass the survival filter, spanning 211 episodes (median two events per contributing episode, median task rate 0.82 rad/s, mean 2.5 contacts). The set includes first certifications after reset and later re-certifications (Fig. 5). The pair score and motion/survival filters select events; hull magnitude is diagnostic, not an additional onset criterion.

Training from the discovered cache. The frozen source supplies one cache row per retained configuration. A fresh teacher trains from this fixed cache with the original task reward, optionally augmented by (4), and the action space, observations, randomization and PPO recipe of Section III. It learns its own actions and is evaluated on the standard cache. Future-survival filtering tests continuation under the moving source; fixed-policy and downstream-training evaluations test resuming motion from zero-velocity resets.

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TABLE II
RESET AND SOURCE-POLICY COMPARISONS ON THE PEN

initialization	rows	survival	twirl	trained sustain (%) $w_{\text{cage}} = 0.1$	trained sustain (%) $w_{\text{cage}} = 0$
standard sweep	897	0.68	1.36	18	5
closure events	561	0.69	1.39	**78**	**71**
random on-policy	684	0.56	1.00	57	55
random on-policy, no filter	800	0.45	0.70	35	20
maximal margin	651	0.63	1.41	26	16
release	533	0.68	1.35	65	35
stationary holds	88	0.15	0.20	20	10
JSRL [22]	–	–	–	40	25
sweep + noise	897	0.55	1.20	18	8
events, 5 % source	254	0.50	1.10	43	25
random, 5 % source	285	0.35	0.70	24	8

Survival: $P(L \geq 390)$ under *cage01*, ten pinned repeats per row; twirl: mean revolutions. Trained: standard-cache sustain after 1800 epochs ($\approx$ 177 M steps), latest checkpoint, 64–160 episodes per run. Last two rows use the first row's 5 % w_{cage}=0 source. –: not applicable; JSRL uses live source execution, without a harvested cache.

B. Control reset sets

Filtered rollout controls. These use the same rollouts and future-survival filter as events. *Maximal-margin* selects up to three positive-ε_6 steps per episode, at least ten steps apart. *Random on-policy* samples up to three steps uniformly, with the same spacing and excluding the first/last five steps. *Stationary holds* require ten consecutive certified steps with $|\omega| < 0.02\,\mathrm{rad/s}$; *releases* use $c_{t-1} > 0, c_t = 0$ after positive rotation. Their retained cache sizes differ (Table II).

Unfiltered resets, diversity and guidance. *Random on-policy, no filter* samples 800 configurations uniformly from the same rollouts without survival filtering and clears velocities: an RSI-style control [17]. *Sweep + noise* perturbs joint positions and object poses from the standard cache, yielding 897 configurations without event selection. *Jump-Start RL* (JSRL) [22] executes the source for the first h training-episode steps, then hands the live state to the fresh policy; h decreases during training. Evaluation uses the fresh policy alone on the standard cache.

C. Fixed-policy viability and downstream training

For the fixed *cage01* teacher, event resets have survival and mean twirl close to the standard cache, and exceed random states on both measures. Release resets perform similarly to onsets despite lacking a certificate at reset; stationary holds perform poorly. Events also transfer to the fixed *cage005* teacher: survival is 0.39 versus 0.38 for the standard cache and 0.22 for random states, although mean twirl is lower than the standard cache (0.22 versus 0.43).

Fresh event-trained teachers average 78% sustain versus 18% from the standard sweep at the same 1800-epoch second-stage budget. Unfiltered random resets reach 35%, while survival-filtered random resets reach 57%, still 21 pp below events. Filtering helps, but filtered random states do not match event selection. Sweep + noise remains at 18%, matching the sweep, while JSRL reaches 40%. Without second-stage shaping, unfiltered random resets, JSRL and noisy sweep starts reach 20%, 25% and 8%, respectively.

Releases and closure onsets have similar fixed-policy survival/twirl (0.68/1.35 versus 0.69/1.39), yet onsets train more effectively (Table II). Without second-stage shaping, events still reach 71% versus 5% for the sweep. This does not establish a shaping-free pipeline: the source teacher used shaping to generate those events.

Pinned evaluation of the trained teachers gives a different ranking: event-trained teachers average 0.55 survival versus 0.11 for sweep-trained teachers, while the random-reset teacher reaches 0.64 and the maximal-margin teacher 0.31. Without shaping, events reach 0.37 versus 0.08. The advantage over random resets therefore concerns conditional sustain and task motion, not every survival metric.

Training cost. Fresh 600-epoch runs ($\approx$ 59 M steps) reach 82% sustain from events, 61% from random on-policy states and 15% from the sweep. Including 1800 source-training epochs gives $1800 + 600 = 2400$ epochs for the event pipeline, compared with 39% sustain after 3600 epochs of single-stage sweep training. This counts training steps; rollout collection and offline processing are additional costs. The 600- and 1800-epoch results are separate runs, not a monotone learning curve.

Events of a failed source. The harvest does not need a source that sustains. The unshaped sweep-trained teacher of Table II ($w_{\text{cage}} = 0$, 5 % sustain) was rolled out 256 times with the training pushes at nominal parameters: 233 rollouts drop and the certificate duty cycle is 0.14, yet they contain 460 onsets, 254 of which pass the survival filter (115 rollouts). Fresh teachers trained from these 254 resets sustain 68, 25 and 37 % over three seeds (mean 43) against 27, 8 and 20 % for the standard sweep with the same seeds (mean 18), and 36 and 13 % without second-stage shaping against 5 %. Random surviving states of the same rollouts give 8, 51 and 13 % (mean 24), so the events beat the sweep in every seed and the random states in two of three. Without second-stage shaping, these random resets reach 8% versus 25% for events. Fixed-teacher survival/twirl is 0.50/1.10 for events and 0.35/0.70 for random resets. A 254-row random subset of the sweep trains to 3 %, the 460 onsets without the survival filter to 8 %, and the uncertified release states to 42 % at 0.25 median revolutions (events: 2.54). The source must catch often enough: a shaped teacher at 8 % sustain (duty 0.11) yields 87 surviving events, which train to 7 %.

Additional fixed-policy object. On a hexagonal pencil outside the ten-object set (7.8 mm across, 180 mm long), 333 events from 98 rollouts yield 0.51 pinned survival and 1.53 mean revolutions over 13 repeats. A 200-row sweep subset gives 0.42/1.17, random on-policy states 0.55/1.13, and maximal-margin states 0.47/1.34. Events produce the most rotation, but random states have the highest survival. This experiment evaluates reset viability, not a downstream training gain.

D. Alternative selection rules

Table III compares ranking by 6-D static margin ε_6, task-oriented margin along T_z [2], and the source teacher's critic

TABLE III

RESET-STATE SELECTION COMPARISONS

selection	pen sustain	pen rev.	pen viab.	cube sustain	egg sustain	mean sustain
ε_6 top-K	28	0.4	0.58	70	50	49
task-GWS top-K	44	1.0	0.64	76	61	60
critic top-K	57	1.5	0.67	81	69	69
closure events	**78**	**2.0**	**0.69**	**86**	**79**	**81**

Sustain: downstream simulation sustain (%), three-seed means; rev.: median task revolutions; viab.: fixed-teacher survival. Mean: unweighted sustain across pen, cube and egg. This comparison is separate from the unequal-size controls of Table II.

TABLE IV

MATCHED-STATE RESET-SELECTION ABLATION

reset selection	motion matched	survival matched	contact matched	event transition	sustain (%)
random rollout	×	✓	×	×	57
motion-matched	✓	✓	×	×	63
motion+contact	✓	✓	✓	×	65
closure event	✓	✓	✓	✓	**78**

value, retaining the top-K states with K set to the event count. Event resets achieve the highest downstream sustain in each object class, exceeding critic selection by 21 pp on the pen, 10 pp on the egg and 5 pp on the cube. These gaps do not alone establish a causal relation between closure duty and learning performance.

E. Matched non-event controls

To isolate the transition criterion from properties of the state itself, we pair each closure event s_e with a non-event state s_r from the same source rollout. Motion-matched controls satisfy

$$|\omega_r - \omega_e| < \delta_\omega, \qquad |V(s_r) - V(s_e)| < \delta_V, \qquad (6)$$

where V is the source critic. The motion-and-contact control additionally requires $|N_{\mathrm{contact},r} - N_{\mathrm{contact},e}| \leq 1$. All controls use the same future-survival filter as the events. Matching motion and critic value raises downstream sustain from 57% to 63%, and additionally matching contact count reaches 65% (Table IV). The true closure-transition resets reach 78%. Thus the gain does not arise merely from selecting states that are already moving, likely to survive, or well contacted; the transition itself contributes information for constructing training resets.

VII. SHAPE GENERALIZATION AND REAL-ROBOT DEPLOYMENT

A. Ten printable objects in simulation

We train object-specific policies with the shaping recipe of Sections III and V. Table V reports best-run sustain and median task revolutions for six compact objects and four elongated or composite objects. Each object has an authored pinch/cradle cache, compact objects start from randomized orientations, and all use $w_{\mathrm{cage}} = 0.1$ with the bounding-cylinder pair score.

TABLE V

OBJECT-SPECIFIC SHAPED TEACHERS IN SIMULATION

object	sustain (%)	median rev.
Compact objects		
ball	98	5.1
rhombohedron	95	3.8
cube	95	4.1
dodecahedron	92	3.5
egg	88	5.2
cylinder	70	3.2
Elongated or composite objects		
pen	98	2.0
dumbbell	90	2.2
marker	88	5.2
chopstick	70	3.2

Best run per object, using its own grasp cache and policy. Revolutions use long-axis twirl for elongated objects and net task-axis rotation for compact objects.

Eight objects reach at least 88% sustain. Among elongated objects with unshaped counterparts, shaping improves marker sustain from 46% to 88% (twirl 1.5 to 5.2), chopstick from 38% to 70% (0.6 to 3.2), and pen from 14% to 98%. These results evaluate a shared shaping rule with separate policies and grasp authoring, rather than a single policy generalizing to unseen objects.

B. Real-robot deployment

Distilled students manipulate eight physical objects (Figs. 1 and 6). We compare certificate shaping, unshaped ($w_{\mathrm{cage}} = 0$), and per-step closure-magnitude shaping. Hardware *Sustain@20s* requires 20 s without a drop, center-height or tilt-bound violation, or one second without net task-direction rotation; the last criterion rejects prolonged stalls.

Fig. 6 reports 50–85% sustain, averaging 68.8% (compact 71.7%; elongated/composite 60.0%), with mean gains of 16.4 pp over unshaped and 10.0 pp over magnitude shaping. Simulation results use a different evaluation protocol.

VIII. LIMITATIONS

Grasp caches and policies are object-specific. Proposition 1 covers ideal planar contacts; estimated proximity contacts, randomized friction and non-cylindrical shapes limit its physical interpretation. Wrench analysis is offline and does not provide a runtime stability guarantee. The ten-object simulation results report best runs. Event discovery requires a source that visits useful catches and uses future survival to filter them; from a failed source the events beat random rollout states in two of three seeds (68/25/37 against 8/51/13%). Noise augmentation and source-guidance results concern the evaluated perturbations and handoff schedule. Source-rollout and evaluation conditions differ; fixed-teacher viability and downstream learning need not rank reset sets alike. The pencil experiment evaluates fixed-teacher resets only, and a second harvesting round produced just 41 events, so iterative improvement remains unestablished. Hardware and simulation sustain use different failure rules.

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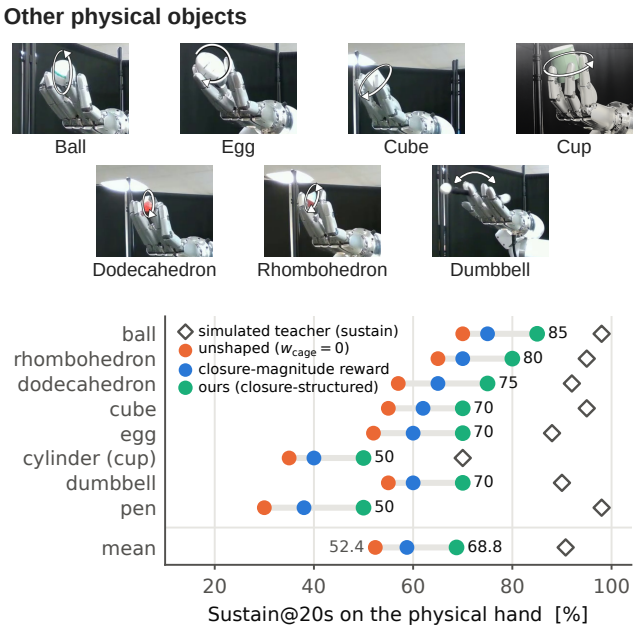


Fig. 6. Other physical objects and hardware Sustain@20s across 160 trials. Double arrows indicate rocking; diamonds show simulated teachers under a different protocol; the bottom row gives object means.

IX. Conclusion

Closure-event grasp discovery selects training resets at transitions into opposing fingertip contact. On the pen, sustain rises from 18% to 78% at equal second-stage budget, compared with unfiltered/filtered random resets (35/57%), Jump-Start RL (40%), matched non-event states (65%) and noisy sweep starts (18%). Events remain useful from a 5%-sustain source. Separately, rotation-gated pair shaping reaches 88% versus 15% for terminal quality and 25% for contact-count rewards. Object-specific shaping and distillation cover ten simulated and eight physical objects. These results support contact transitions as a source of training states, within the tested source-policy and evaluation settings.

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